

SANJANA SINGH

Data Science & AI

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SUMMARY

A recent USF computer science graduate with hands-on experience utilizing AI and data science to solve real-world issues, such as creating an LLM-powered application and evaluating NLP using N gram and GPT-based models. I leverage Python and Tableau to conduct EDA, data visualization, and machine learning, effectively transforming raw datasets into actionable insights. My background in data science, statistics, and machine learning is reinforced by my credentials from Google and Villanova. I led more than 500 event applications and oversaw multi-channel interactions as the Marketing Chair for the Google Developer Group at USF. I am a results-oriented, early-career professional with strong technical skills, communication abilities, and a proactive approach to achieving goals.

EXPERIENCE

AI-Powered SOP Redundancy Detection Agent

Jun 2026

- Developed an LLM agent that analyzes Global SOPs and Site SOPs to identify semantically redundant procedures and policy statements within GMP-regulated documentation. Implemented prompt engineering and semantic similarity workflows to evaluate document overlap based on contextual meaning rather than exact text matching. Built an interactive web application allowing users to upload SOP documents, run automated comparisons.
- Benefit/Outcome:** AI Agent driven solution reduces manual document review process, improves efficiency in SOP document maintenance, and opportunity for future standardization and streamline compliance review process.
- Technologies:** Python, Streamlit, OpenAI API GPT model

Video to Training Document Pipeline

Jun 2026

- Built a multi-stage AI pipeline converting MP4 screen recordings into formatted Word training documents. Architected six sequential GPT-4o stages (frame extraction, UI change detection, action recognition, step aggregation, document writing, QA review), each passing structured JSON to the next. Deployed as a Streamlit web app with live progress tracking and one-click download.
- Benefit/Outcome:** Eliminates manual documentation effort for tutorial and onboarding content; output passes automated QA review before delivery, ensuring accuracy and reducing human review time.
- Technologies:** Python, Streamlit, OpenAI API GPT-4o, OpenCV, python-docx

N-Gram Language Model Evaluation

Mar 2026

- Designed a controlled degradation study to evaluate how well unigram, bigram, and trigram models detect grammatical violations using perplexity. Trained models on 55k Brown Corpus sentences, constructed an 800-sentence evaluation dataset with four degradation levels, and applied Mann-Whitney U statistical tests to validate findings. Extended analysis to DistilGPT-2 output to explore lightweight hallucination detection.
- Benefit/Outcome:** Demonstrated that bigram/trigram perplexity can serve as a lightweight hallucination detection signal for LLM pipelines.
- Technologies:** Python, NLTK

LEADERSHIP AND EXTRACURRICULARS

Marketing Lead

Mar 2025 - May 2026

Google Developer Group at USF

- Led marketing for DevFest Tampa and HackUSF, driving 500+ applications, 280+ participants, and 85+ project submissions across Florida.
- Increased HackUSF applications from ~200-300 to 500+ through digital campaigns, campus outreach, and partnerships.
- Designed, filmed, and edited promotional media content, while capturing high-quality event photography to drive organic reach and sponsor highlight campaigns.
- Partnered with cross-functional teams, corporate sponsors, and university stakeholders to elevate event attendance and community engagement.

SKILLS

</> Programming & AI

Python, C, LLM (OpenAI API), Natural Language Processing (NLP), Prompt Engineering, SQL



Data Science & ML

Pandas, Matplotlib, Seaborn, Scikit-learn, NLTK, Jupyter Notebook, Tableau



Tools & Platforms

VS Code, Streamlit, Microsoft Excel, PowerPoint, Word, Teams



Relevant Coursework

Software Engineering, Image Processing Fundamentals, Augmented Reality, Data Storage & Analysis

EDUCATION

Bachelor of Science in Computer Science

University of South Florida (USF), College of Engineering — Tampa, FL

CERTIFICATES

Google Advanced Data Analytics

- Foundations of Data Science
- Go Beyond the Numbers: Translate Data into Insights
- The Power of Statistics
- Regression Analysis: Simplify Complex Data Relationships
- The Nuts and Bolts of Machine Learning
- Google Advanced Data Analytics Capstone

Villanova University

- Certificate in Artificial Intelligence and Machine Learning